



The Twin Transition: How Digitalisation Supports Sustainable SMEs

🕒 15 min · Supporting learning resource (LSM)

Module 1 · Understanding the Green Economy

Unit 4 · The Twin Transition: Digitalisation and Sustainability



Introduction

Small and medium-sized enterprises are currently facing two major transformations: the green transition and the digital transition. These are not separate developments. When combined strategically, digitalisation can significantly support environmental sustainability and improve business performance.

This Learning Support Material complements the core lesson by explaining the concept of the twin transition and showing how digital tools can support sustainability goals in SMEs. It also highlights the potential risks of digitalisation and the importance of responsible digital use.

By reading this material, you will understand how digital and environmental actions can reinforce each other and how these concepts relate to your own organisational context.

Key concepts explained

The Twin Transition: Digital and Green Together

The twin transition refers to the parallel and interconnected shift toward:

- A sustainable, low-impact economy
- A digitally enabled business environment

Digital technologies can help SMEs monitor resource use, optimise processes, and improve environmental performance. However, digital growth must align with sustainability objectives.

The core idea is simple: digitalisation should accelerate environmental improvement—not create additional environmental burdens.

How Digitalisation Supports Sustainability

Digital tools enable SMEs to make better, data-driven decisions.

Examples include:

- Monitoring energy consumption in real time
- Tracking material flows
- Automating production processes to reduce waste
- Using data analytics to optimise logistics
- Implementing digital reporting tools for environmental performance

These solutions help businesses identify inefficiencies and reduce unnecessary resource use.

For SMEs, even small digital improvements can produce measurable environmental and financial benefits.

Risks and Responsible Digital Use

Digitalisation is not automatically sustainable.

Potential risks include:

- Increased energy consumption from digital infrastructure
- Growth of electronic waste
- Overly complex digital systems
- Dependence on energy-intensive cloud services
- Lack of digital skills within the organisation

Responsible digitalisation means:

- Choosing energy-efficient technologies
- Extending equipment life cycles
- Avoiding unnecessary digital expansion
- Training employees in responsible digital practices
- Aligning digital investments with sustainability goals

The twin transition requires strategic thinking, not only technological adoption.

Practical examples or typical situations

- A small manufacturing SME installs smart meters to monitor electricity consumption and identifies peak usage patterns, reducing energy costs.
- A service-based SME replaces paper-based processes with digital documentation, reducing material use and improving efficiency.
- A logistics SME uses route optimisation software to lower fuel consumption and reduce emissions.

These examples demonstrate how digital solutions can directly support environmental performance.

Link to next unit

In the next unit, we explore how sustainability and competitiveness converge in SMEs.

The focus will shift to ecological resilience, strategic positioning, and how responsible environmental practices become a source of long-term competitive advantage.

Why this matters for SMEs

The integration of digital and sustainability strategies affects SMEs in several ways:

Costs:

Digital monitoring can identify energy and material waste, reducing operating expenses.

Efficiency:

Automation and data analytics improve productivity and process optimisation.

Risk management:

Transparent data supports regulatory compliance and reduces exposure to environmental risks.

Competitiveness:

Digitally mature and environmentally responsible SMEs are more attractive to customers, partners, and investors.

However, digital investments should be aligned with real business needs. Over-investment or poorly planned digitalisation can increase costs without delivering environmental benefits.

Key takeaways

- The twin transition connects digitalisation and sustainability.
- Digital tools can improve environmental performance in SMEs.
- Data enables better resource management decisions.
- Digitalisation carries environmental and operational risks.
- Responsible digital use strengthens competitiveness and resilience.

Reflection prompts

- Which digital tools are currently used in your organisation?
- Do they contribute to environmental efficiency?
- Could digital monitoring improve resource management?
- Are there hidden environmental costs linked to your digital systems?
- How can digital investments align better with sustainability goals?